

# Arun Vignesh Malarkkan

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## About Me

- 📖 Investigating reliability and generalization of LLM reasoning, combining causal inference and mechanistic interpretability, focusing on invariance discovery, spurious correlation mitigation, and robust evaluation.

## Education

- Aug 2023 – Present     📖 **Ph.D., Computer Science**, Arizona State University, Tempe, AZ.  
Advisor: Dr. Yanjie Fu. Knowledge Discovery and Data Mining Lab.  
CGPA: 3.9/4.0.
- May 2020     📖 **Master of Science, Computer Science**, Arizona State University, Tempe, AZ.  
CGPA: 3.8/4.0.

## Technical Skills

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| Core Expertise              | 📖 LLM Reasoning & Reliability, Recommendation Systems, Causal Inference & Causal Representation Learning, Mechanistic Interpretability, Post-training & Fine-tuning, Retrieval-Augmented Generation (RAG), Data-Centric AI. |
| ML & Systems                | 📖 PyTorch, TensorFlow, Hugging Face Transformers, Distributed Training, GPU/CUDA Programming.                                                                                                                               |
| LLM & Generative AI         | 📖 PEFT (LoRA, QLoRA), LangChain, LlamaIndex, OpenAI API, vLLM.                                                                                                                                                              |
| Interpretability & Analysis | 📖 TransformerLens, Activation Patching, Probing, PyTorch Hooks.                                                                                                                                                             |
| Programming                 | 📖 Python, Java, TypeScript.                                                                                                                                                                                                 |
| Data & Infrastructure       | 📖 AWS, GCP, BigQuery, MySQL, Apache Spark (SparkML), Docker.                                                                                                                                                                |
| General Libraries           | 📖 Scikit-learn, Pandas, NumPy, Flask, AutoGluon.                                                                                                                                                                            |

## Research Publications

### First-Author Publications

- 1     **A. V. Malarkkan**, U. Byahut, M. R. Choudhury, Y. Charde, V. Gupta, and Y. Fu, “Invariant reasoning directions in latent trajectories of language models,” in *ICML Workshop on Mechanistic Interpretability*, 2026.
- 2     **A. V. Malarkkan**, X. Wang, K. Liu, D. Zhang, and Y. Fu, “Causally-guided diffusion for stable feature selection,” 2026, arXiv preprint. Under review, KDD.
- 3     **A. V. Malarkkan**, W. Ying, and Y. Fu, *CAFE: Causally-guided automated feature engineering with multi-agent reinforcement learning*, arXiv preprint. Under review, NeurIPS, 2026.
- 4     **A. V. Malarkkan** et al., *FinRule-Bench: A benchmark for joint reasoning of llms over financial tables and principles*, arXiv preprint. Under review, EMNLP, 2026.
- 5     **A. V. Malarkkan** et al., *Latent reasoning does not lack signal, it lacks invariance: Subspace-constrained refinement for robust latent reasoning*, arXiv preprint. Under review, NeurIPS, 2026.
- 6     **A. V. Malarkkan**, H. Bai, A. Kaushik, and Y. Fu, “DELTA: Variational disentangled learning for privacy-preserving data reprogramming,” in *Proceedings of the IEEE International Conference on Data Mining (ICDM)*, 2025.
- 7     **A. V. Malarkkan**, H. Bai, D. Wang, and Y. Fu, “Causal graph profiling via structural divergence for robust anomaly detection in cyber-physical systems,” in *Proceedings of the IEEE International Conference on Big Data*, 2025.

- 8 A. V. Malarkkan, D. Wang, H. Bai, and Y. Fu, "Incremental causal graph learning for online cyberattack detection in cyber-physical infrastructures," *IEEE Transactions on Big Data*, 2025.
- 9 A. V. Malarkkan, D. Wang, and Y. Fu, "Multi-view causal graph fusion based anomaly detection in cyber-physical infrastructures," in *Proceedings of the ACM International Conference on Information and Knowledge Management (CIKM)*, 2024.

## Co-Authored Publications

- 1 H. Cao et al., *Not all correct gradients should be trusted: Shortcut-aware reasoning via directional detection and correction*, arXiv preprint. Under review, NeurIPS, 2026.
- 2 H. Cao et al., *Robust data reshaping via flat and consensus-seeking diffusion*, arXiv preprint. Under review, NeurIPS, 2026.
- 3 X. Wang, K. Liu, A. V. Malarkkan, and Y. Fu, *Evolving demonstration optimization for chain-of-thought feature transformation*, arXiv preprint. Under review, KDD, 2026.
- 4 W. Ying et al., *Distribution shift aware neural tabular learning*, arXiv preprint. Under review, ICML, 2026.
- 5 W. Ying et al., *A survey on data-centric ai: Tabular learning from reinforcement learning and generative ai*, arXiv preprint., 2025.

## Professional Experience

- May 2026 – Present     **Research Intern**, The Home Depot (Recommendation Systems).
- Translating frontier LLM post-training research into production product compatibility systems that verify, at catalog scale, whether products can be correctly used or installed together.
  - Applying constrained preference optimization and post-training alignment methods to make compatibility predictions robust and consistent under noisy real-world product data, paraphrased queries, and catalog drift.
  - Designing evaluation protocols that stress-test compatibility reasoning for reliability under input variation and distribution shift, bridging research-grade robustness analysis with deployment requirements.
- Jun 2025 – Present     **Doctoral Researcher**, DOW Chemicals (National Academy of Engineering Grant).
- Leading research on incorporating LLM-guided knowledge priors into neuro-symbolic, causally-aware multi-agent reinforcement learning frameworks to encode domain constraints and guide structured reasoning in materials discovery.
  - Developing frameworks that integrate causal structure with sequential decision-making to improve robustness under distribution shifts in industrial simulation environments.
  - Designing interpretable and scalable models for material optimization under uncertainty (details subjected to confidentiality).
- Jan 2023 – Jun 2023     **Senior Software Engineer**, Fidelity Investments.
- Led development of Monte Carlo-based financial projection systems for retirement planning at scale.
  - Built probabilistic simulation frameworks for uncertainty-aware decision modeling in long-term financial forecasting.
  - Developed ensemble-based outlier detection models to identify key drivers affecting projected savings outcomes.

## Professional Experience (continued)

- Jun 2021 – Jan 2023     **Software Development Engineer**, Amazon — Alexa AI.
- Developed and deployed purchase likelihood modeling for Alexa recommendation systems, improving voice-driven purchases by +8%.
  - Built end-to-end decision and ranking pipelines for voice-based commerce, integrating user behavior modeling with real-time inference systems.
  - Designed production systems for voice-enabled transactional flows with compliance constraints (e.g., parental consent, HIPAA-aligned systems).
- Jun 2020 – May 2021     **Data Scientist / Software Developer**, ASU Decision Center for Educational Excellence.
- Developed predictive models for demographic, financial, and infrastructure planning using large-scale public datasets.
  - Designed end-to-end data pipelines (AWS, ETL) for data ingestion, modeling, and deployment.
  - Built models for resource allocation and policy analysis, identifying key drivers of graduation rates and community-level outcomes.

## Academic Research Experience

- 2024 – Present     **LLM Reasoning: Causal Mediation and Invariant Mechanisms**
- Investigating whether LLMs encode causally invariant reasoning mechanisms or rely on spurious correlations, using interventional analysis over Transformer computations.
  - Developed a causal mediation framework over intermediate tokens, combining activation patching and counterfactual inputs to estimate mediated effects on model predictions, demonstrating that only a sparse subset of tokens act as stable causal mediators.
  - Identified a mechanistic view of reasoning as a multi-stage process: candidate formation, causal mediation, redundancy, and invariant consolidation.
- 2025 – 2026     **FinRule-Bench: Benchmarking LLM Reasoning under Rule Constraints** (Under review — KDD 2026)
- Developed a benchmark for evaluating rule-based reasoning and diagnostic completeness in LLMs over structured financial data.
  - Introduced causal-counterfactual evaluation protocols to assess reasoning consistency and failure modes.
- 2024 – 2026     **Causally-Guided Automated Feature Engineering with Multi-Agent Reinforcement Learning** (Under review — ICML 2026)
- Developed a causal-RL framework for feature construction using causal graphs as inductive priors.
  - Designed structured exploration and reward shaping for robustness under distribution shift.
- 2024 – 2025     **Privacy-Preserving Generative Data Reprogramming** (Accepted — ICDM 2025)
- Proposed a variational generative framework with causal disentanglement for privacy-aware feature transformation.
  - Achieved ~9% performance improvement and ~35% reduction in privacy leakage.
- 2023 – 2025     **Research on Causal Learning and Robustness**
- Conducted research on causal representation learning and invariant modeling for improving robustness under distribution shift, focusing on disentangling stable mechanisms from spurious correlations across learning systems.
  - Developed incremental and multi-view causal graph learning frameworks for robust modeling under distribution shifts and streaming environments (IEEE TBD, IEEE Big Data, CIKM).
  - Focused on invariant structure learning and continual adaptation in non-stationary systems.

## Teaching, Mentorship & Academic Service

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### Invited Talks

Dec 2025

■ **Keynote** — IEEE International Conference on Modelling, Simulation & Intelligent Computing (MoSICom), BITS Pilani, UAE.

*"Beyond Correlation: Can Causal Reasoning Enhance Modern AI Models in Robustness and Interpretability?"*

Presented research on causal inference, robustness, and interpretability in modern AI systems, emphasizing limitations of correlation-based learning and the role of causal reasoning.

### Teaching

Aug 2023 – Present

■ **Senior Teaching Assistant / Lab Instructor**, Arizona State University.

- Led instruction and lab sessions for courses in Object-Oriented Programming, Data Mining, and Semantic Web Mining for large undergraduate cohorts, and also project management for 30 Capstone Project teams.
- Designed and developed hands-on lab modules and course materials, emphasizing applied machine learning and data-driven reasoning.

### Mentorship

Jan 2024 – Present

■ **Research Mentor**, Arizona State University.

- Mentoring undergraduate and graduate researchers on Neuro-symbolic AI for feature engineering, world model simulations, and LLM reasoning optimization.
- Guiding projects on investigating efficient fine-tuning and evaluation of large language models for reasoning tasks, including experimental design and model analysis.

### Academic Service

Jan 2025 – Present

■ **Program Committee Member**: AAAI 2026; IEEE ICSCAN 2024; IEEE BigData 2024 (Undergraduate & High School Research Symposium); IEEE ICDM 2025.

**Conference Reviewer**: KDD, ICLR, NeurIPS, ICDM, IEEE BigData, Urban AI.

**Journal Reviewer**: ACM Transactions on Knowledge Discovery from Data (TKDD).

**Session Chair**: IEEE MoSICom 2025; IEEE ICDM 2025 — Undergraduate & High School Research Symposium.

### Workshops & Outreach

■ **Outreach**

- Conducted hands-on workshops on Machine Learning fundamentals and Graph Neural Networks, training undergraduate students in applied AI techniques.
- Developed instructional material focused on practical model development and real-world applications.
- IEEE Phoenix Chapter Student Volunteer & Organizer

### Honors & Awards

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2025

■ **IEEE ICDM Scholarship**, IEEE International Conference on Data Mining, 2025.

2018

■ **Academic Scholarship**, Arizona State University, 2018.

2017–2018

■ **Best Undergraduate Researcher Award**, 2017–2018.